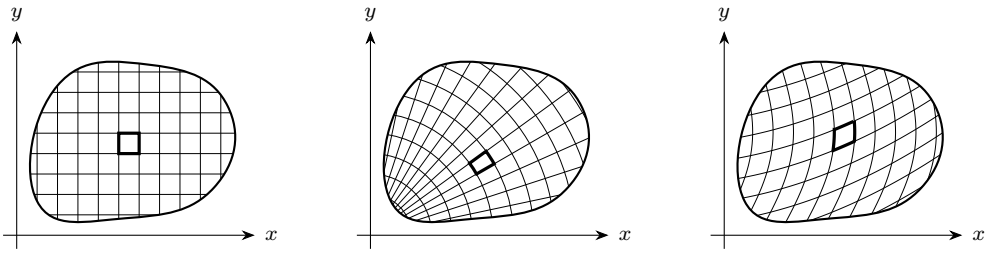
We Have Done This Before

When we switched a double integral to polar coordinates, we cut the region into pieces using a different grid. In Cartesian coordinates the grid lines are $x =$ constant (vertical lines) and $y =$ constant (horizontal lines), and one small piece has area $\Delta A = \Delta x\,\Delta y$. In polar coordinates the grid lines are $r =$ constant (circles) and $\theta =$ constant (rays), and one small piece has area $\Delta A \approx r\,\Delta r\,\Delta\theta$. That extra factor of r is the reason we write $dA = r\,dr\,d\theta$.

Today we ask the same question for *any* pair of new variables u and v. The grid curves $u =$ constant and $v =$ constant can be any two families of curves. What is the area of one small piece? Whatever it is, that factor is what we tack onto the end of the new integral.



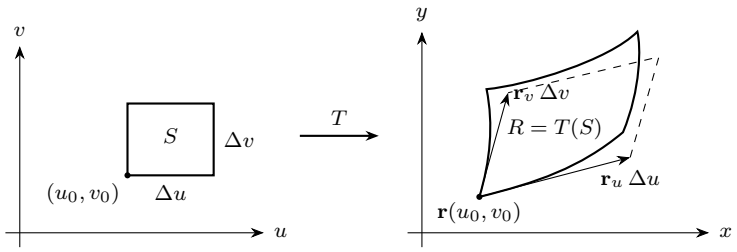
Transformations

A change of variables is a **transformation** $T(u, v) = (x, y)$, given by two formulas

$$x = g(u, v), \qquad y = h(u, v).$$

T takes a region S in the uv-plane to a region $R = T(S)$ in the xy-plane. We want to trade an integral over R (the hard region) for an integral over S (an easy region, usually a rectangle).

How much does T stretch area? Take a tiny rectangle in S with sides Δu and Δv. Its image in R is a slightly curved shape. Write the transformation as the vector function $\mathbf{r}(u, v) = g(u, v)\,\mathbf{i} + h(u, v)\,\mathbf{j}$. The sides of the curved shape are approximately the tangent vectors $\mathbf{r}_u\,\Delta u$ and $\mathbf{r}_v\,\Delta v$, and those two vectors make a parallelogram.



The area of a parallelogram is the magnitude of a cross product, so

$$\Delta A \approx \left\|\mathbf{r}_u\,\Delta u \times \mathbf{r}_v\,\Delta v\right\| = \left\|\mathbf{r}_u \times \mathbf{r}_v\right\|\Delta u\,\Delta v, \qquad \mathbf{r}_u \times \mathbf{r}_v = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial x}{\partial u} & \frac{\partial y}{\partial u} & 0 \\ \frac{\partial x}{\partial v} & \frac{\partial y}{\partial v} & 0 \end{vmatrix} = \left(\frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}\right)\mathbf{k}.$$

Only the $\mathbf{k}$-component survives, and it is a 2×2 determinant. That determinant gets a name.

Definition (Jacobian). The **Jacobian** of the transformation $T(u,v) = \big(g(u,v), h(u,v)\big)$ is

$$J(u,v) = \frac{\partial(x,y)}{\partial(u,v)} = \begin{vmatrix} \dfrac{\partial x}{\partial u} & \dfrac{\partial x}{\partial v} \\ \dfrac{\partial y}{\partial u} & \dfrac{\partial y}{\partial v} \end{vmatrix} = \frac{\partial x}{\partial u}\frac{\partial y}{\partial v} - \frac{\partial x}{\partial v}\frac{\partial y}{\partial u}.$$

Small areas scale by $|J|$: $\Delta A \approx |J(u,v)|\,\Delta u\,\Delta v$.

Change of Variables for Double Integrals. If T maps S one-to-one onto R, then

$$\iint_R f(x,y)\,dA = \iint_S f\big(g(u,v), h(u,v)\big)\left|\frac{\partial(x,y)}{\partial(u,v)}\right| du\,dv.$$

Substitute for x and y, change the region from R to S, and tack on the absolute value of the Jacobian.

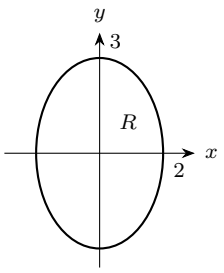
Example 1. Find the Jacobian of the polar transformation $x = r\cos\theta$, $y = r\sin\theta$.

Example 2. Use the transformation $x = 2u$, $y = 3v$ to find the area of the region R inside the ellipse $9x^2 + 4y^2 \le 36$.

(a) Sketch the region S in the uv-plane.

(b) Find the Jacobian.

(c) Find the area of R using Area $= \displaystyle\iint_R 1\,dA$.



Practice 1. Use the transformation $x = 5u$, $y = 2v$ to find the area of the region R inside the ellipse $4x^2 + 25y^2 \leq 100$.

When u and v Are Given in Terms of x and y

Often the integrand or the boundary of R suggests u and v as functions of x and y, such as $u = x + y$ and $v = x - y$. To find $\partial(x, y)/\partial(u, v)$ we would first have to solve for x and y. There is a shortcut: the two Jacobians are reciprocals.

Reciprocal Jacobian.

$$\frac{\partial(x,y)}{\partial(u,v)} = \frac{1}{\dfrac{\partial(u,v)}{\partial(x,y)}}, \qquad \text{where} \qquad \frac{\partial(u,v)}{\partial(x,y)} = \begin{vmatrix} \dfrac{\partial u}{\partial x} & \dfrac{\partial u}{\partial y} \\ \dfrac{\partial v}{\partial x} & \dfrac{\partial v}{\partial y} \end{vmatrix}.$$

If the answer still has x and y in it, rewrite it in terms of u and v.

The Routine.

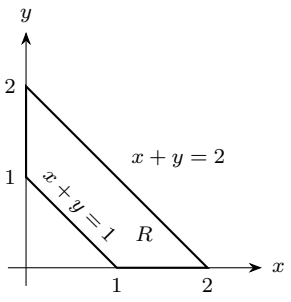
1. Choose u and v. Look at the integrand and at the boundary curves of R.
2. Find S: rewrite each boundary curve of R in terms of u and v, then sketch.
3. Find the Jacobian $\partial(x, y)/\partial(u, v)$, using the reciprocal if that is easier.
4. Rewrite the integral over S, with $|J|\,du\,dv$ on the end, and evaluate.

Example 3. Evaluate $\iint_R \cos\left(\frac{y-x}{y+x}\right) dA$, where R is the trapezoid with vertices $(1,0)$, $(2,0)$, $(0,2)$, and $(0,1)$. Use $u = y + x$ and $v = y - x$.

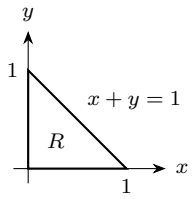
(a) Write each side of R in terms of u and v, and sketch S.

(b) Find the Jacobian.

(c) Evaluate the integral.



Practice 2. Evaluate $\iint_R \left(\frac{x-y}{x+y}\right)^4 dA$, where R is the triangle bounded by $x + y = 1$, $x = 0$, and $y = 0$. Use $u = x - y$ and $v = x + y$.

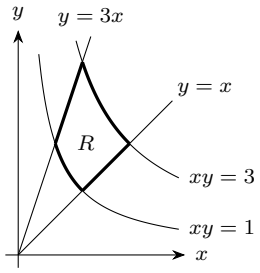


Example 4. Evaluate $\iint_R xy\,dA$, where R is the region in the first quadrant bounded by the lines $y = x$ and $y = 3x$ and the hyperbolas $xy = 1$ and $xy = 3$. Use $u = xy$ and $v = y/x$.

(a) Write each boundary of R in terms of u and v, and sketch S.

(b) Find the Jacobian. Write your answer in terms of u and v.

(c) Evaluate the integral.



Answers

Example 1. $\dfrac{\partial(x,y)}{\partial(r,\theta)} = \cos\theta\,(r\cos\theta) - (-r\sin\theta)\sin\theta = r$. This is the r in $dA = r\,dr\,d\theta$. (In three dimensions, the same idea produces the $\rho^2\sin\varphi$ in spherical coordinates.)

Example 2. S is the unit disk $u^2 + v^2 \le 1$; $J = 6$; Area $= \iint_S 6\,du\,dv = 6\pi$.

Practice 1. S is the unit disk; $J = 10$; Area $= 10\pi$.

Example 3. S: $1 \le u \le 2$, $-u \le v \le u$; $\dfrac{\partial(u,v)}{\partial(x,y)} = -2$, so $\left|\dfrac{\partial(x,y)}{\partial(u,v)}\right| = \dfrac{1}{2}$;

$$\int_1^2\int_{-u}^{u} \cos\left(\frac{v}{u}\right)\frac{1}{2}\,dv\,du = \frac{1}{2}\int_1^2 2u\sin 1\,du = \frac{3\sin 1}{2}.$$

Practice 2. S: $0 \le v \le 1$, $-v \le u \le v$; $|J| = \dfrac{1}{2}$; $\displaystyle\int_0^1\int_{-v}^{v} \frac{u^4}{v^4}\cdot\frac{1}{2}\,du\,dv = \frac{1}{10}$.

Example 4 (if not finished in class). S: $1 \le u \le 3$, $1 \le v \le 3$; $\dfrac{\partial(u,v)}{\partial(x,y)} = \dfrac{2y}{x} = 2v$, so $|J| = \dfrac{1}{2v}$;

$$\int_1^3\int_1^3 u\cdot\frac{1}{2v}\,du\,dv = \left(\int_1^3 u\,du\right)\left(\int_1^3 \frac{1}{2v}\,dv\right) = 4\cdot\frac{\ln 3}{2} = 2\ln 3.$$